GIT QUEST | 30-DAY GIT LEARNING SERIES

Day 2: git init and First Commit

Week 1: First Steps | Level: Absolute Beginner

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Today's mission

Yesterday you learned that Git saves your work and GitHub shares it. Today you create your first Git project and make your first save point from the terminal.

- Turn any folder into a Git repository with `git init`
- Check what Git sees with `git status`
- Make a commit with `git add` and `git commit`
- See your save point with `git log`

1. The big idea: opening a save slot

A game needs a **save slot** before it can store anything. `git init` creates that slot for a folder by adding a hidden folder called `.git`. That folder is the **vault** where every commit will live.

> **Remember this one-liner**
> Run **git init** once per project. After that, **git add** and **git commit** again and again.

2. Step by step

Step 1: Create a project folder

```
$ mkdir my-first-repo
$ cd my-first-repo
```

Step 2: Turn it into a repository

```
$ git init -b main
Initialized empty Git repository in /home/you/my-first-repo/.git/
```

Step 3: Create a file

```
$ echo "Hello, Git!" > hello.txt
```

Step 4: Ask Git what it sees

```
$ git status
On branch main

No commits yet

Untracked files:
        hello.txt
```

Step 5: Pick the file for saving

```
$ git add hello.txt
```

Step 6: Make your first commit

```
$ git commit -m "Add hello.txt"
[main (root-commit) 3f2a9c1] Add hello.txt
 1 file changed, 1 insertion(+)
```

Step 7: See your save point

```
$ git log --oneline
3f2a9c1 (HEAD -> main) Add hello.txt
```

> **Tip**
> `-b main` names your first branch *main* and needs Git 2.28 or newer. On older versions run `git init` then `git branch -M main`.

3. Cheat sheet

Command	What it does
git init -b main	Creates a new repository in the current folder
git status	Shows what Git sees right now
git add <file>	Picks a file for the next save
git commit -m "message"	Makes the save point with a short message
git log --oneline	Lists your save points, one per line

Common problems

You see	Fix
fatal: not a git repository	cd into your project folder, or run git init
Author identity unknown	Run the two git config --global lines from Day 1
nothing to commit	Create or edit a file, then git add it

4. Today's 3 commits

Create these files in your repo, then add and commit each one. You can use the GitHub web editor or your new terminal skills.

#	File	Content	Commit message
1	day02-first-commit/README.md	Today's lesson with steps and cheat sheet	docs(day02): add lesson on git init and first commit
2	day02-first-commit/demo.tape	VHS script that records the terminal GIF	feat(day02): add VHS tape for git init demo
3	day02-first-commit/exercise.md and main README.md	Challenge, and the progress tracker moves to 2/30	chore(day02): add exercise and update progress tracker

5. Optional: record the terminal GIF with VHS

Save this as `demo.tape`, run `vhs demo.tape`, and commit the tape and the GIF.

```
Output demo.gif
Set FontSize 20
Set Width 1000
Set Height 580
Set Theme "Dracula"
Type "git init -b main"
Enter
Sleep 2s
Type 'echo "Hello, Git!" > hello.txt'
Enter
Type "git add hello.txt"
Enter
Type 'git commit -m "Add hello.txt"'
Enter
Sleep 4s
```

6. Mermaid diagram GitHub draws for you

````
```mermaid
gitGraph
  commit id: "first commit"
```
````

7. Challenge and quiz

Challenge: make a new repo with `about-me.txt`, commit it, edit the file, and commit again. `git log --oneline` should show 2 commits.

Q1. Which command turns a folder into a Git repository?

Q2. What does "untracked" mean in git status?

Q3. What is the hidden folder that stores your commits called?

Answers: 1) git init. 2) Git sees the file but you haven't added it yet. 3) .git

> **Coming up: Day 3**
> **The 3 areas of Git.** Learn the working directory, staging area and repository, and why git add exists.